

Week 1

Keel Bone Chicken Project

01/16/2026



College of Engineering

This Week

This week's progress

- **PCB Reordering**
 - Purchase Request sent to Procurement Office
 - Initial Purchase Failed + New Purchase
- **PCB Design**
 - Updates to Rev. 4 of PCB Design
 - Added Force + Strain Logging
 - Several Design Choices to Minimize Area
- **Data Analysis**
 - Able to determine chicken movement from PCB
 - Need to change some firmware on the Nicla Sense
- **This Semester**

PCB Reordering

Purchase Request sent to Procurement Office

- Drill files regenerated and issue dealt with with JLCPCB support
- Need to call procurement office to ensure the order successfully goes through
- May be an issue with the resistors
 - 2 mm (length) x 1 mm (height) sizing too small for regular PCBA
 - May need to switch to 4 mm (length) x 2 mm (height) for our PCB

PCB Reordering

Initial Purchase Failed + New Purchase Successful

- Order successfully went through
- Card never charged, this time drill files are present and card will be charged

support@jlcpcb.com
+86 755 23919769
JLCPCB.COM

Batch No.: W2025122002394691
Ship Via: DHL Express (DDP)
Type of Trade: DDP

Ship To:

Purdue Polytechnic Institute
Nathan Huang
363 North Grant Street
Indiana WEST LAFAYETTE 47907
US
Email:huangnathan69@gmail.com
Tel:+13176567438
VAT No:

Billing To:

Purdue Polytechnic Institute
Nathan Huang
363 North Grant Street
Indiana WEST LAFAYETTE 47907
UNITED STATES OF AMERICA
Email:huangnathan69@gmail.com
Tel:3176567438
VAT No:

	Product	File Name	Order Number	QTY	Unit Price	Ext.Price
1	Rigid Populated printed circuit board	Gerbers2_Y5	SMT025121962393-Y5	10	USD \$7.23	USD \$72.30

Merchandise Total:	USD \$72.30
Shipping:	USD \$50.10
Subtotal:	USD \$122.40
States Sales&Use Tax:	USD \$8.57
Import Taxes :	USD \$32.54
Grand Total:	USD \$163.51

Tax rates vary for different products; please refer to the [tax details](#).

PCB Design

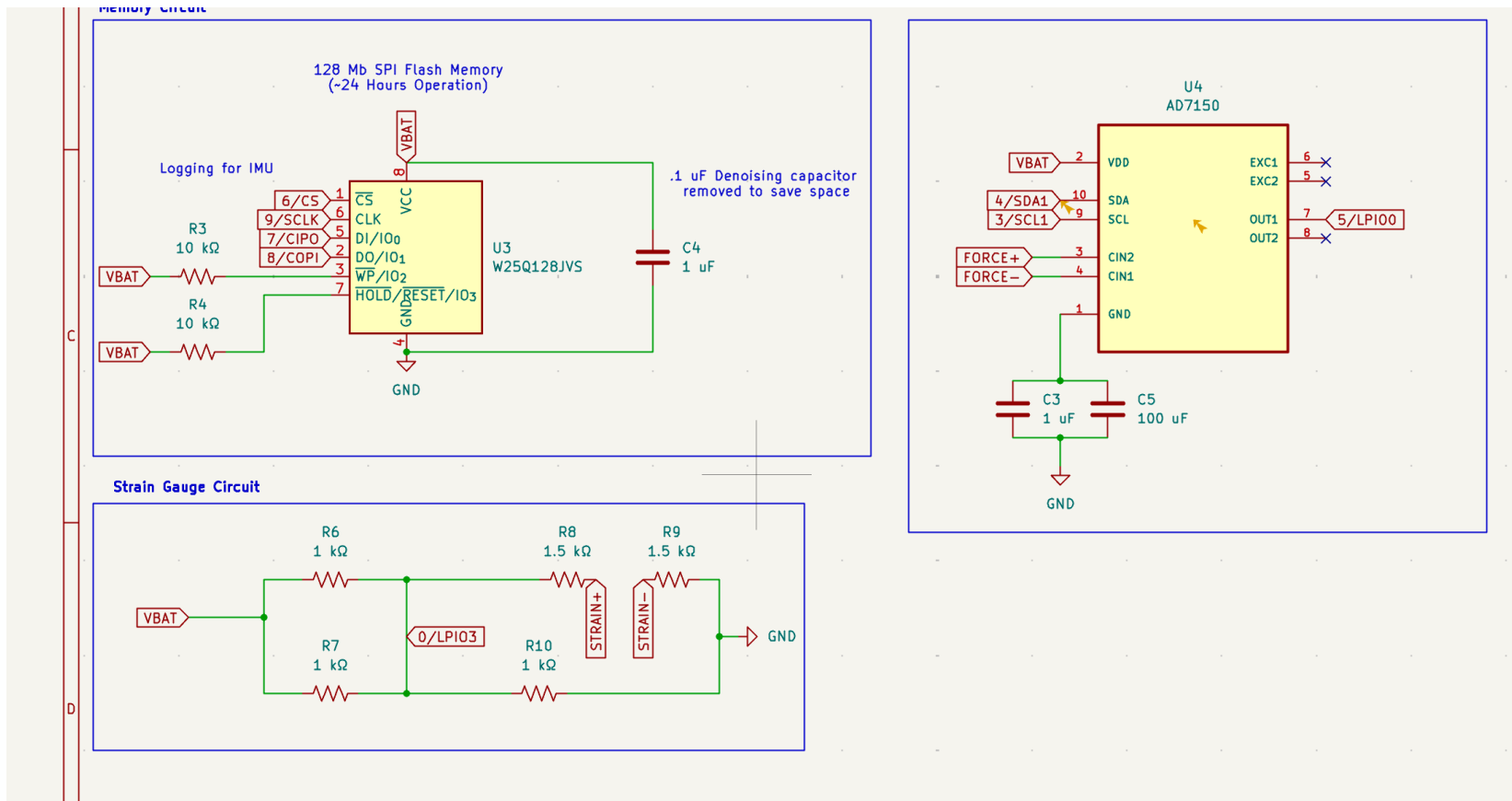
Updated Rev. 4 of PCB Design

- Design Files + Footprints found and generated
- Footprint of added hardware is smaller than anticipated, which is good
- Minimal Issues with Routing this time around

PCB Design

Added Force + Strain Logging

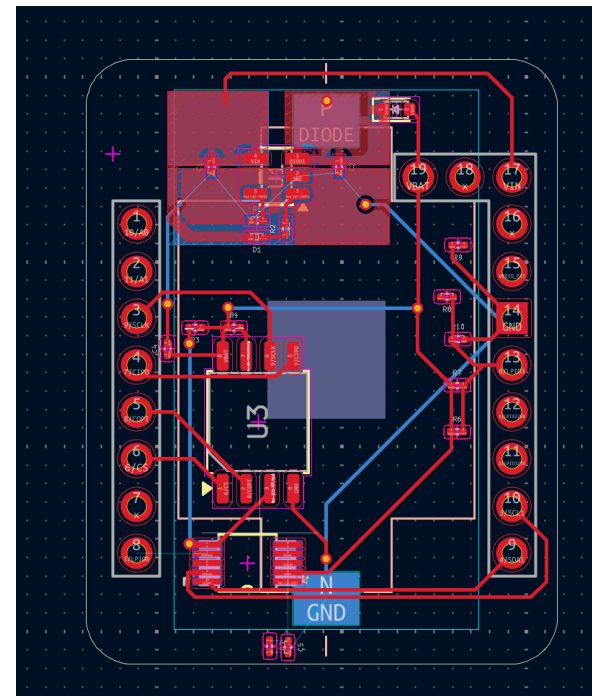
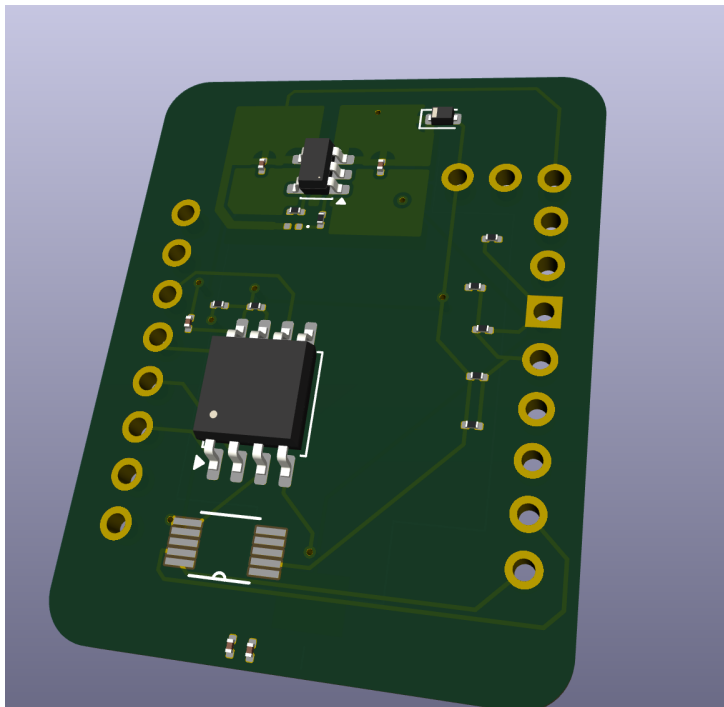
- Sensor Circuitry can be seen in the Schematic below...
- Added to PCB on the next slide



PCB Design

Several Design Choices to Minimize Area

- Second Smallest Resistors Chosen due to Assembly Issues
- Took precautions to minimize taken area
 - Space needed for Ribbon/Flat Flex Cables in the Future
 - Space needed for power dissipation in the future



Data Analysis

Able to determine chicken movement from PCB

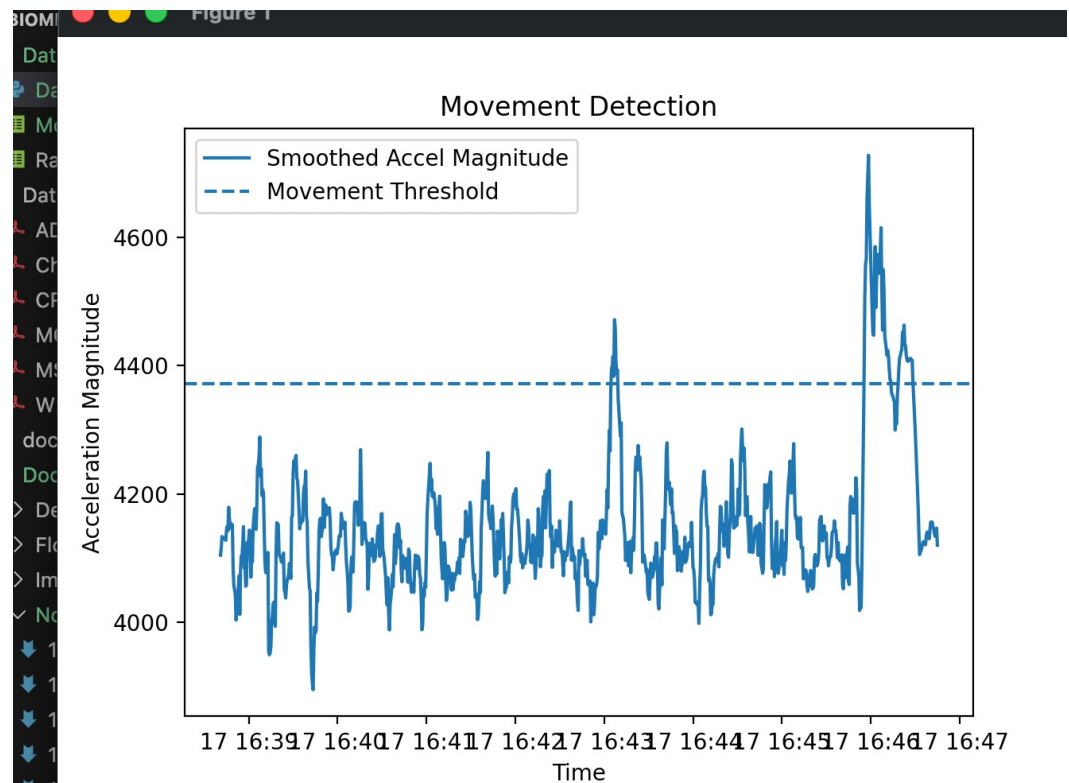
- From the PCB, ~16 minutes 37 seconds of data was logged
 - I had originally set the maximum entries to 1000 -> 16:40
 - Missing 3 seconds of data is good, likely startup time
 - From the data, I was able to run the following data smoothing program:

```
Data > DataProcessing.py > ...
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 PATH_NAME = "Movement1.csv"
6
7 df = pd.read_csv(PATH_NAME)
8
9 df["timestamp"] = pd.to_datetime(df["timestamp"])
10
11 df["accel_mag"] = np.sqrt(
12     df["accelerometer_Ax"]**2 +
13     df["accelerometer_Ay"]**2 +
14     df["accelerometer_Az"]**2
15 )
16
17 # Rolling Window
18 df["accel_mag_smooth"] = df["accel_mag"].rolling(
19     window=10, center=True
20 ).mean()
21
22 # Movement threshold
23 threshold = [
24     df["accel_mag_smooth"].mean() +
25     2 * df["accel_mag_smooth"].std()
```


Data Analysis

Need to change some firmware on the Nicla Sense

- IMU seems to turn off after extended periods of time on battery (20+ minutes)
 - Need to look into this issue and see how it can be avoided in the future
 - From data collection, I was able to obtain the following plot
 - Very visible when the chicken choose to move vs. fly



Next Week

Blockers & Next Week's Work

- Need to find the suitable battery replacement on the PCB
- Work on Hybrid Approach for Data Collection
- Fix Issue where Nicla IMU turned off unexpectedly after a prolonged time of data collection
- Get Sample Data on Force Impacts (Small, Medium, Large Impacts)
- Prepare the 2+ hour version of data test

This Semester

Outline Entire Semester Goals

Questions:

Thank You

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